Fertility and Housing Tenure Choice

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Since May

One-market lifecycle model of housing and fertility. What changed:

- **Sequential fertility**: birth-by-birth attempts under declining fertility; children ever born and children at home tracked separately
- **Idiosyncratic income risk**: persistent after-tax earnings process, set externally
- **One housing market**: the two-location structure is set aside for the national question

Model

Environment

- Households live J four-year periods from age 18; retirement, then survival risk
- Two child states: n children ever born, m children currently at home
  - Children at home raise consumption and housing needs
  - Each child matures stochastically, leaves the household, and enters the economy as a young adult
- Idiosyncratic persistent earnings risk; wages and the interest rate are exogenous
- Households rent, or own a discrete house size on a ladder financed with a mortgage
- Government: property tax rebated lump sum; payroll tax funds a pension
- New households enter as young renters: matured children plus an outside inflow

Preferences

Flow utility over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{1}{1-\sigma}\left[\frac{c^{\alpha}\left(\mathbf{h} - \underline{h}\, m\right)^{1-\alpha}}{e(m)}\right]^{1-\sigma} + \psi_{\text{child}}\, m.$$

- **Space requirement.** Each child at home absorbs $\underline{h}$ rooms before housing yields services
- **Equivalence scale.** $e(0) = 1$ and $e' > 0$: a larger family needs more resources to reach the same utility
- **Owner premium.** $\mathbf{h} = \chi h$ for owners, with $\chi > 1$; $\mathbf{h} = h^R$ for renters

Bequests

Warm-glow utility at death over terminal wealth:

$$\mathcal{B}(\widetilde{b}) = \theta_0 \frac{(\theta_1 + \widetilde{b})^{1-\sigma}}{1-\sigma}, \qquad \widetilde{b} = b + ph.$$

- θ_0 governs the strength of the bequest motive; θ_1 makes bequests a luxury
- Estates are valued by parents but transfer to no one: no intergenerational wealth flow

Fertility and Child Maturation

- Between ages 18 and 45, households decide each period whether to try for a child: a discrete choice with taste shocks, with separate scales κ_f before and after the first child
- Trying yields a birth with the conception probability

$$\pi(\text{age}) = 1 - \omega_1\, e^{\omega_2(\text{age}-18)}, \qquad \pi = 0 \text{ after } 45,$$

  which falls from 0.98 at 18 to 0.62 at 40: postponed births are increasingly likely never to happen
- Each child lives at home for 18 years on average, with stochastic maturation, raising the equivalence scale and the space requirement
- Matured children enter the economy as young households

Firms and Earnings

- A competitive firm produces with labor, so the wage is pinned down by technology and the labor market clears at any aggregate efficiency units L:

$$Y = A\,L, \qquad w = A$$

- Households supply efficiency units inelastically: an age profile times a persistent idiosyncratic component

$$\log y_{j+1} = \rho \log y_j + \varepsilon_{j+1}, \qquad \varepsilon \sim N(0, \sigma_\varepsilon^2)$$

- Earnings are taxed progressively, a payroll tax funds a stationary pension, and a property tax on owners is rebated lump sum
- With constant returns, a larger population leaves the wage unchanged: housing is the only fixed factor

Housing, Tenure, and Budget Constraints

Households rent a continuous quantity up to a cap, or own a discrete size on a ladder:

$$h^R \in (0, \bar{h}^R], \qquad h \in \{H_1, \dots, H_K\}.$$

Tenure segmentation: $\max H_k > \bar{h}^R$, so family-sized space is reachable only through ownership.

Renters

$$c + b' + r\, h^R = (1+q)\, b + y(j), \qquad b' \geq 0.$$

Owners, with prior stock h_{-1}

$$c + b' + (\delta + \tau_H)\, ph + \mathbf{1}\{h \neq h_{-1}\}\big[ph - (1-\psi)\, ph_{-1}\big] = (1+q)\, b + y(j), \qquad b' \geq -\phi\, ph.$$

- q: interest rate; δ: maintenance; τ_H: property tax; ψ: transaction cost; $1-\phi$: down-payment share

Housing Supply and Market Clearing

Housing supply slopes upward in levels, and rents map into prices through user cost:

$$H^S = H_0 \, r^{\xi}, \qquad r = (q + \delta + \tau_H) \, p.$$

- One market-clearing condition: aggregate housing services demanded equal H^S
- Supply is in levels, not per household: a permanently larger population permanently raises rents
- Stationary equilibrium: the price, the transfer, and the distribution are constant, and the government budget clears

The Household's Problem

Within the period, a household first decides whether to try for a child, then chooses tenure, housing, and saving:

$$V_j(b,h,y,n,m) = \mathbb{E}_\varepsilon \max_{a\in\{0,1\}} \Big\{ W_j^a(b,h,y,n,m) + \varepsilon_a \Big\},$$

$$W_j^a = \max_{\text{rent/own},\, h',\, h^{R\prime},\, b'} \; u(c,\mathbf{h};m) + \beta\,\mathbb{E}\big[V_{j+1}(b',h',y',n',m')\big]$$

subject to the renter or owner budget and the collateral constraint; a successful try ($a=1$, probability π_j) adds a child next period.

In old age there are no fertility decisions; survival risk and the warm glow at death close the problem.

Population Closure

Closing the Population

Every version of this model needs a rule for the size of the entering cohort. Three ways to supply one:

1. **Balanced growth path.** Population and quantities grow at a common rate and are read per capita. The committed housing requirement makes preferences non-homothetic and the housing stock is fixed in levels, so no such path exists here. It is also counterfactual: the economy is not observed growing along one
2. **Full dynamic equilibrium.** Solve the transition after the reform, with prices clearing each period. Expensive to compute, and it is no longer clear what the calibration should target, since the observed moments would have to be assigned to a date on the path
3. **Stationary equilibrium.** Compare stationary states before and after. This delivers a single number, at the cost of a rule that pins the population level

The stationary comparison is used here.

The Entry Rule

Both protocols keep the same accounting. They differ in what pins the inflow M:

$$\text{earlier:} \quad M = M(V) \qquad\qquad \text{now:} \quad M = \bar{M}.$$

- **Earlier.** Entry rises when the economy becomes more attractive. The strength of that response, $\lambda = \partial \ln M / \partial V$, is identified by no moment, and it drives the answer: population responses came out three to ten times larger
- **Now.** The quota fixes M, so policy moves the population only through births, and the rent response damps it
- **Cost.** Migration cannot respond to a more attractive country. The earlier numbers are kept as a bound on that channel

Population Balance

Entrants each period are retained matured children plus the immigration inflow M:

$$\underbrace{S\,E_0}_{\text{entrants needed}} \quad = \quad \underbrace{R\,S\,B_0}_{\text{retained matured children}} \quad + M,$$

with E_0 the replacement flow per household, B_0 the matured-children flow, and R the share who stay.

Given the fixed inflow, the stationary population

$$S = \frac{M}{E_0 - R\,B_0}$$

exists and is stable: as the population shrinks, a fixed inflow becomes a growing share, and the decline stops.

R and M are pinned at the baseline by two observables: the outside-origin share of young entrants and the population normalization.

Population in Counterfactuals

With R and M fixed, population responds to policy through births alone. Differentiating the balance condition:

$$\%\Delta S \;=\; \frac{1 - s^{out}}{s^{out}}\,\%\Delta B_0,$$

where s^{out} is the immigrant share of entrants. Each immigrant household founds a family line that fades out at the native replacement rate; a policy that raises births makes every line last longer.

At the measured share (16.9% of young entrants, ACS), the multiplier is about five. Housing pushes the other way: a larger population raises rents, which discourages births.

Calibration

Externally Calibrated Parameters

	Object	Value	Source
Pref.	σ risk aversion	2.0	standard
Fin.	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
Life	17 four-year periods	ages 18–85	retirement 65
Income	ρ, σ_ε, HSV τ	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
Children	maturation hazard	2/9 per period	18 expected years
	fecundity ω_1, ω_2	0.02, 0.134	conception schedule
Housing	ladder, rental cap	$\{2, \dots, 11\}$, 6 rooms	room units

External restriction on the search: annual $\beta \geq 0.94$.

Internally Calibrated Parameters

Param	Description	Estimate	Bounds	Near bound?
β (annual)	Discount factor	0.9946	$[0.94, 0.9995]$	no
ψ_{child}	Child flow utility	0.000	$[0, 3]$	**at lower**
κ_f	Attempt taste scale, first birth	2.987	$[0.02, 50]$	no
κ_f^{cont}	Attempt taste scale, later births	1.713	$[0.02, 50]$	no
χ	Owner service premium	1.037	$[0.1, 5]$	no
H_0	Housing supply intercept	61.706	$[1.25, 500]$	no
θ_0	Bequest strength	0.159	$[0, 8]$	**near lower**
θ_1	Bequest shifter	0.094	$[0.02, 16]$	**near lower**
$\underline{h}$	Child room floor	0.614	$[0.1, 1.8]$	no

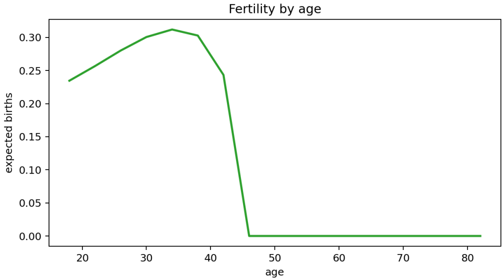
Nine parameters, twelve moments. ψ_{child} is estimated at zero, and the bequest parameters sit near their lower bounds.

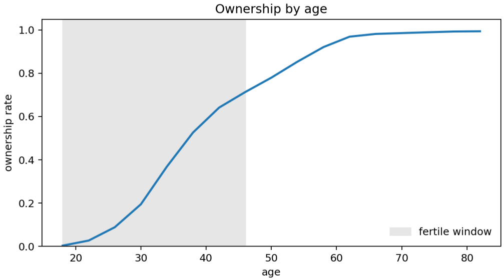
Model Fit

	Moment	Target	Model	Gap	Contr.
Fertility	Completed fertility	1.918	1.967	+0.049	3.4
	Childlessness	0.188	0.104	−0.084	**121.5**
	Mean age at first birth	25.31	25.57	+0.26	1.1
	First births at 30+	0.270	0.327	+0.057	**51.5**
Housing	Rooms response to first birth	0.664	0.690	+0.026	0.6
	Rooms, 3+ vs. 1–2 children	0.368	0.359	−0.009	0.2
	Family ownership gap	0.168	0.162	−0.006	0.5
	Ownership rate	0.575	0.582	+0.007	0.1
	Aggregate mean rooms	5.780	6.478	+0.698	5.8
Wealth	Wealth / earnings	6.873	5.794	−1.079	7.3
	Bequest flow / wealth	0.0088	0.0087	−0.0001	0.0
	Old wealth p90/p50	3.448	2.090	−1.358	**105.1**

The remaining misses are childlessness, the dispersion of old-age wealth, and the share of late first births; additional search does not improve them.

Lifecycle Profiles





Policy Results

Funded Property-Tax and Grant Reforms

Raise the property tax from 1% to 2%; rebate lump-sum, or fund a purchase grant for family-sized homes. All changes vs. the funded 1% baseline, percent:

Arm	Policy	TFR	Births	Househ.	$-\Delta$Imm.	Price	Rent
Floor	2% tax	+0.38	+0.33	+1.70	1.74	−17.5	−6.1
Floor	tax + grant	+0.75	+0.67	+3.46	3.52	−17.3	−5.8
Tilt	2% tax	+0.20	+0.18	+0.90	0.92	−17.7	−6.3
Tilt	tax + grant	+0.49	+0.44	+2.23	2.26	−17.2	−5.7

In U.S. units, +0.33–0.67% of births is roughly 12–24 thousand additional babies per year. The tax lowers the asset price, so the down payment falls by 17.5%, while renter rents — which include the tax — fall by 6%.

Reading the Population Numbers

The same number can be read three ways:

- more households in the long run, at a fixed immigration inflow;
- fewer immigrants required to hold the population fixed;
- the share of the fertility gap closed by the policy.

Housing feedback is present but small: the demographic arithmetic overstates the response by at most a tenth of a percentage point.

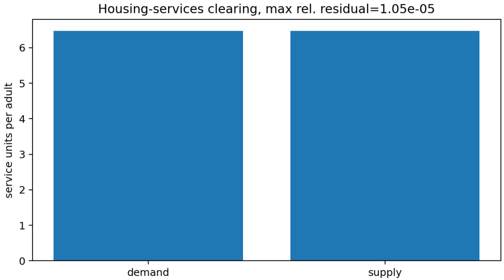
Timing: the birth and immigration flows move immediately, while the household count converges over generations — half of the adjustment takes roughly a century.

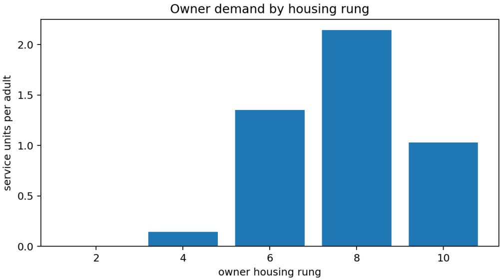
Open Decisions

1. **Anchoring the inflow.** Among household heads aged 18–22, recent immigrants are only 3.7% — but immigrants mostly arrive older than the model's entry age. Either anchor the inflow to total net migration in entrant equivalents (roughly 17–22%, close to the 16.9% in use), or model entry at several ages
2. **The fertility block.** The closure requires natives to replace at least 83% of each cohort; the current fit sits close to that edge, and the childlessness and old-age wealth misses are limitations of the mechanism rather than of the search
3. **Welfare.** With different populations across counterfactuals, welfare is compared per household

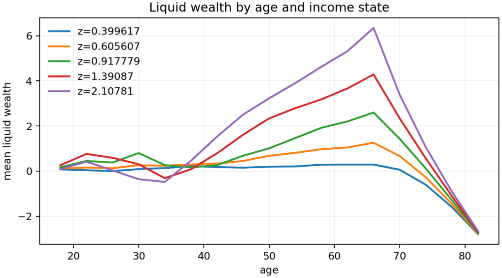
Appendix

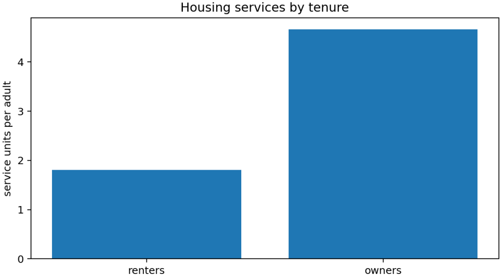
Appendix: Housing Market Diagnostics



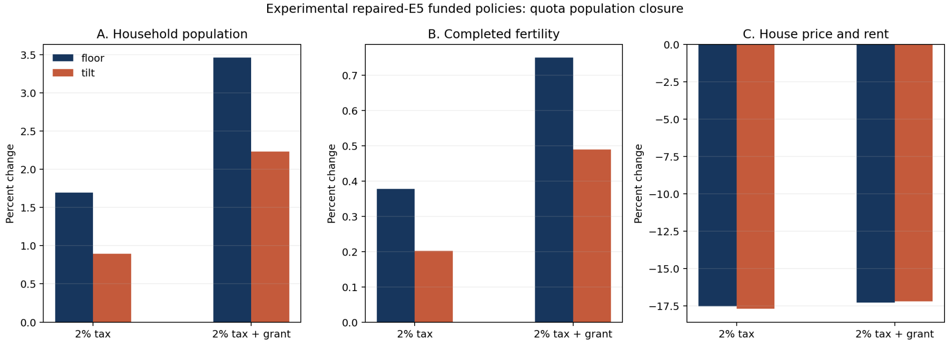


Appendix: Wealth and Tenure Diagnostics

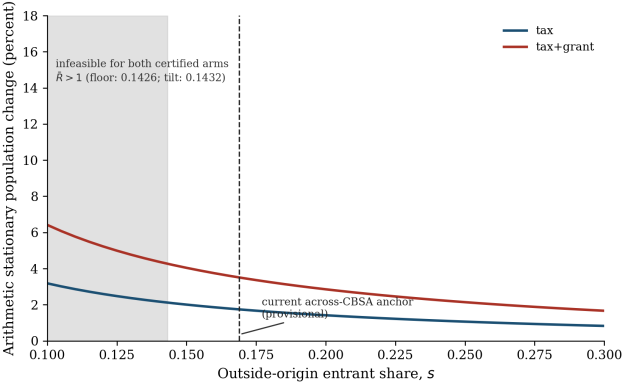




Appendix: Policy Comparison



Appendix: Anchor Sensitivity and Convergence Speed



Stationary population response as a function of the outside-origin share s^{out}, with the infeasible region ($s^{out} < 0.143$) shaded. Convergence: generational decay $\rho = \bar{R}B_0/E_0 = 0.831$ gives a half-life of 97–112 years ($g = 26$–30); an age-structured variant with uniform birth timing gives ~ 230 years (upper bound).

Appendix: The Retired Logit as a Sensitivity Row

The old protocol's numbers survive as a labeled robustness row, not a model:

Arm	Policy	Quota population	Logit sensitivity ($\lambda = 2$, unidentified)
Floor	2% tax	+1.70	+9.98
Floor	tax + grant	+3.46	+10.10
Tilt	2% tax	+0.90	+8.99
Tilt	tax + grant	+2.23	+9.06

The gap between columns is what an unidentified taste parameter was contributing. The logit row bounds "what if entry responded to values" without pretending to identify it.

Appendix: National Entry-Anchor Measurement (ACS 2012–2023)

Measure (household heads unless noted)	Share	Multiplier $(1-s)/s$
Ages 18–22, foreign-born, arrived $\leq$ 4y (headline)	3.71%	25.9
Ages 18–22, foreign-born, arrived $\leq$ 8y	5.36%	17.7
Ages 18–25, foreign-born, arrived $\leq$ 4y	3.47%	27.9
Ages 18–22, foreign-born stock (upper bound)	9.04%	10.1
Persons 18–22, foreign-born, arrived $\leq$ 4y	2.89%	33.7
Quota feasibility floor	14.3%	—
Age-aggregated net-migration bracket (to verify)	17–22%	3.5–4.9

Every entry-age-window measure is infeasible; the age-aggregated concept is not. The two are different objects because immigrants mostly arrive above the stylized entry age — the anchor decision on the Open Decisions slide.

Appendix: Multi-Start Record (overnight)

16 fresh chains, unchanged 12-moment / 9-parameter contract, 225-minute budget each, strict-repeat certification:

	Best	Median	Worst
Fresh chains (seeds 09–24)	263.5	$\sim$284	370.5
Certified incumbent (Aug 5, chain 6)	249.186 — **stands**		

All 16 eligible with exact strict repeats on the run's best; market residuals $\leq 2.2 \times 10^{-5}$. With 24 chains total across the two arrays, the remaining misses are search-converged.